

Taxonomic Gap Analysis

GBIF Gap Analysis - Species Coverage & Conservation Priorities

gbifgaps

February 11, 2026

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Overview

This report analyzes **taxonomic gaps** in the national biodiversity data, comparing coverage against the national backbone and examining threatened species representation.

Key Questions:

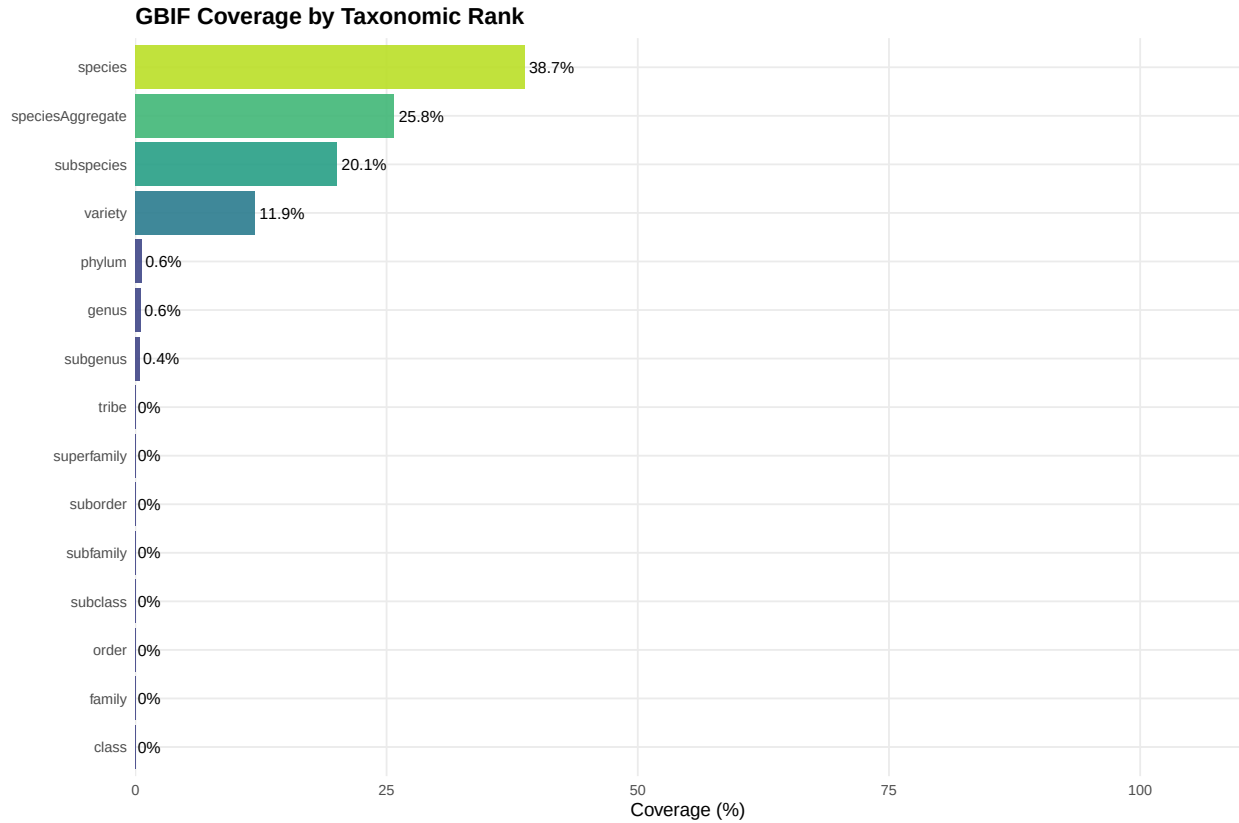
- What proportion of the country's known species are in GBIF?
- Which taxonomic groups have the best/worst coverage?
- How well are threatened species represented?
- Which species should be prioritized for data collection?

1. Overall Taxonomic Coverage

1.1 Summary Statistics

```
## - **Taxa in backbone reference**: 168,043
## - **Taxa found in GBIF**: 51,525
## - **Overall coverage**: 30.7%
## - **Priority taxa identified**: 1,351
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1.2 Coverage by Rank



1.3 Orders by Kingdom: Backbone vs GBIF Coverage

This section shows all orders in the backbone grouped by class, where:

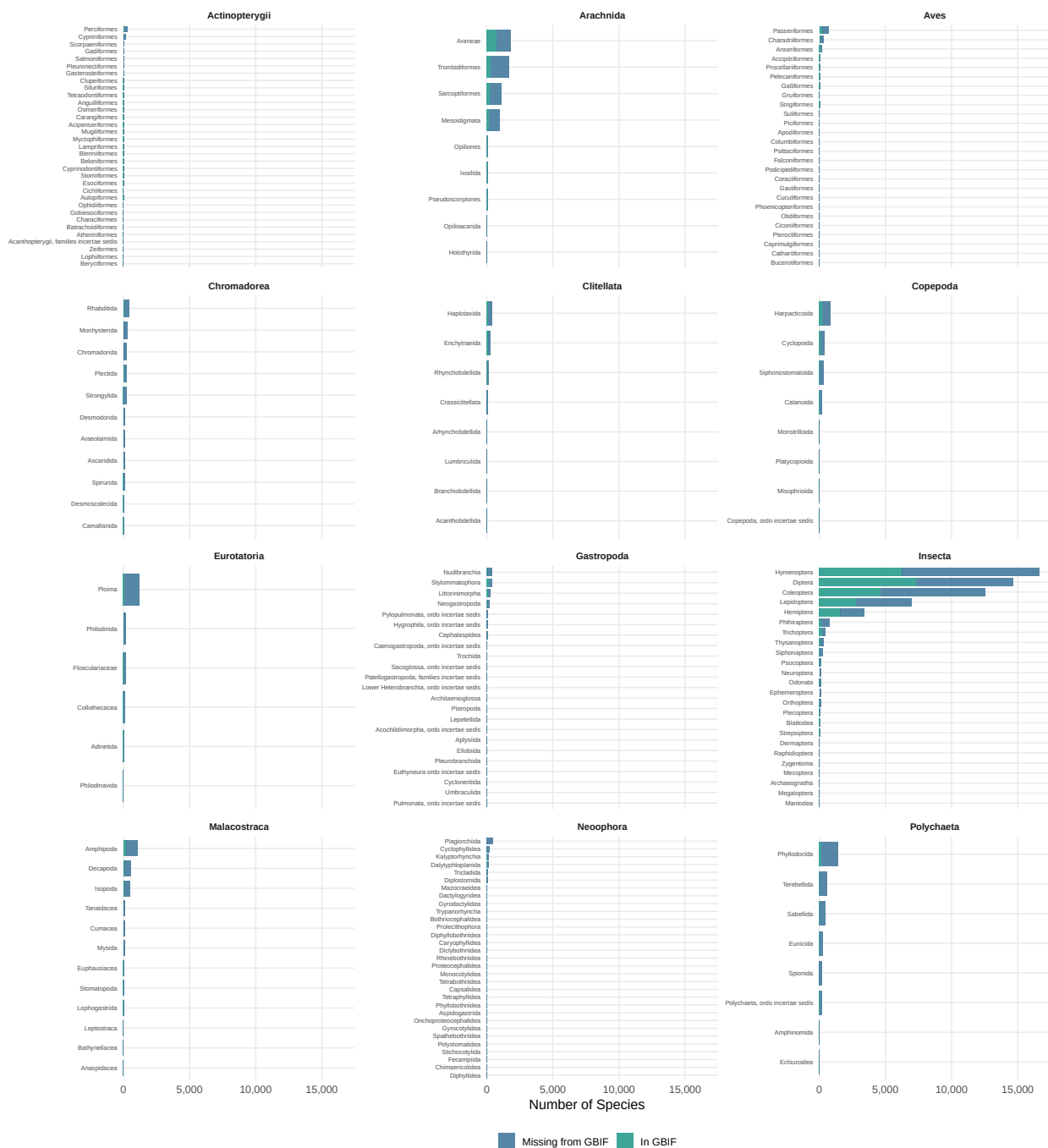
- **Bar length** = total number of species in the backbone for that order
- **Green portion** = species found in GBIF
- **Blue portion** = species missing from GBIF

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## **Total orders in backbone**: 1093
## **Total species in backbone**: 166,599
## **Species found in GBIF**: 51,522 (30.9%)
```

Animalia

Animalia: Species Coverage by Order (Absolute)

Bar length = species in backbone | Green = in GBIF | Blue = missing



Animalia: Species Coverage by Order (Percentage)

All orders = 100% of backbone species | Green = % in GBIF | Blue = % missing

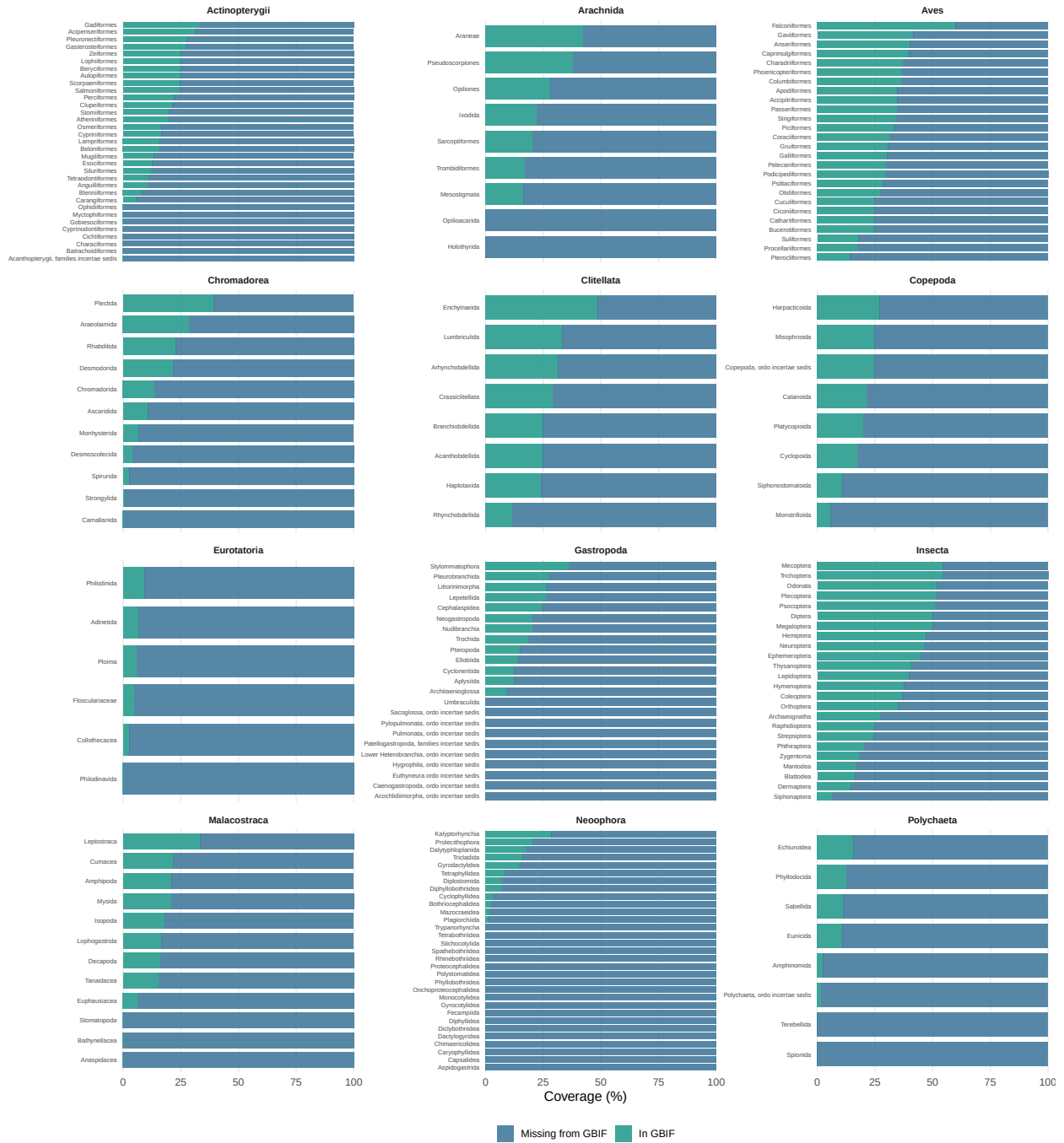


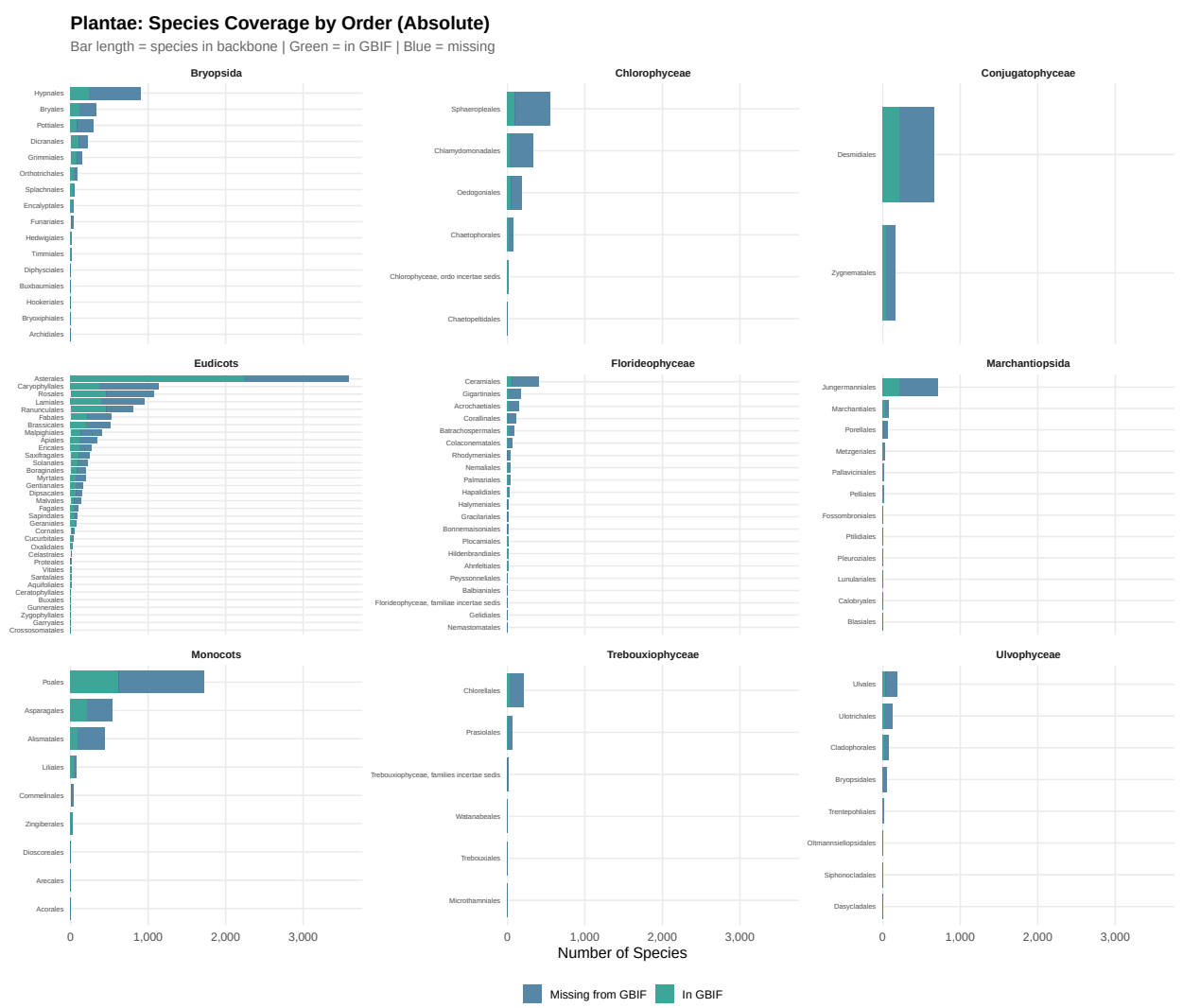
Table 1: Animalia: Species Coverage by Class

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Insecta	24	57032	23595	33437	41.4
Arachnida	9	5787	1502	4285	26.0
Polychaeta	8	3166	267	2899	8.4
Malacostraca	12	2661	502	2159	18.9
Chromadorea	11	2036	342	1694	16.8

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Gastropoda	23	1922	382	1540	19.9
Aves	26	1817	625	1192	34.4
Copepoda	8	1787	375	1412	21.0
Neophora	31	1641	116	1525	7.1
Eurotatoria	6	1617	95	1522	5.9
Actinopterygii	33	1065	216	849	20.3
Clitellata	8	901	275	626	30.5
Entognatha	3	878	285	593	32.5
Enoplea	5	771	67	704	8.7
Hydrozoa	10	743	133	610	17.9
Bivalvia	14	700	173	527	24.7
Ostracoda	4	677	76	601	11.2
Demospongiae	15	564	96	468	17.0
Gymnolaemata	2	456	91	365	20.0
Mammalia	9	444	115	329	25.9
Branchiopoda	8	311	66	245	21.2
Eutardigrada	2	256	26	230	10.2
Anthozoa	8	237	48	189	20.3
Acoelomorpha, classis incertae	2	191	56	135	29.3
sedis					
Diplopoda	7	188	53	135	28.2
Palaeacanthocephala	2	149	13	136	8.7
Asteroidea	6	134	22	112	16.4
Thecostraca	6	131	25	106	19.1
Chondrichthyes	10	122	19	103	15.6
Ophiuroidea	2	113	15	98	13.3
Chilopoda	4	109	35	74	32.1
Holothuroidea	4	109	14	95	12.8
Polycladidea	3	101	17	84	16.8
Stenolaemata	1	99	24	75	24.2
Hoplonemertea	2	94	21	73	22.3
Sipunculidea	2	89	7	82	7.9
Amphibia	2	76	15	61	19.7
Pilidiophora	1	73	15	58	20.5
Cephalopoda	6	71	16	55	22.5
Heterotardigrada	2	70	7	63	10.0
Entoprocta, ordines incertae sedis	1	64	16	48	25.0
Palaeonemertea	4	63	4	59	6.3
Echinoidea	4	61	13	48	21.3
Macrostromorpha	3	57	17	40	29.8
Pycnogonida	1	57	9	48	15.8
Calcarea	2	53	11	42	20.8
Polyplacophora	2	41	12	29	29.3
Paupoda	1	36	7	29	19.4
Scyphozoa	3	33	7	26	21.2
Phylactolaemata	1	28	4	24	14.3
Sagittioidea	2	26	4	22	15.4
Scaphopoda	2	24	5	19	20.8
Enteropneusta	1	23	0	23	0.0
Articulata	1	22	4	18	18.2
Gordioida	3	22	2	20	9.1
Solenogastres	3	19	0	19	0.0

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Archiacanthocephala	2	18	0	18	0.0
Crinoidea	1	18	2	16	11.1
Tentaculata	2	18	5	13	27.8
Phascolosomatidea	2	17	1	16	5.9
Staurozoa	1	15	3	12	20.0
Symphyla	1	15	0	15	0.0
Caudofoveata	1	13	3	10	23.1
Eoacanthocephala	1	13	1	12	7.7
Inarticulata	1	13	1	12	7.7
Ichthyostraca	2	12	2	10	16.7
Priapulida, classis incertae sedis	2	12	2	10	16.7
Leptocardia	1	10	0	10	0.0
Nuda	1	9	3	6	33.3
Petromyzontida	1	9	3	6	33.3
Phoronida, classis incertae sedis	1	9	0	9	0.0
Pararotatoria	1	8	0	8	0.0
Xenoturbellida, classis incertae sedis	1	7	0	7	0.0
Graptolithoidea	1	6	0	6	0.0
Eucyclophora	1	5	0	5	0.0
Homoscleromorpha	1	5	0	5	0.0
Merostomata	1	4	0	4	0.0
Myxini	1	4	1	3	25.0
Polypodiozoa	1	3	0	3	0.0
Tantulocarida	1	2	0	2	0.0

Plantae



Plantae: Species Coverage by Order (Percentage)

All orders = 100% of backbone species | Green = % in GBIF | Blue = % missing

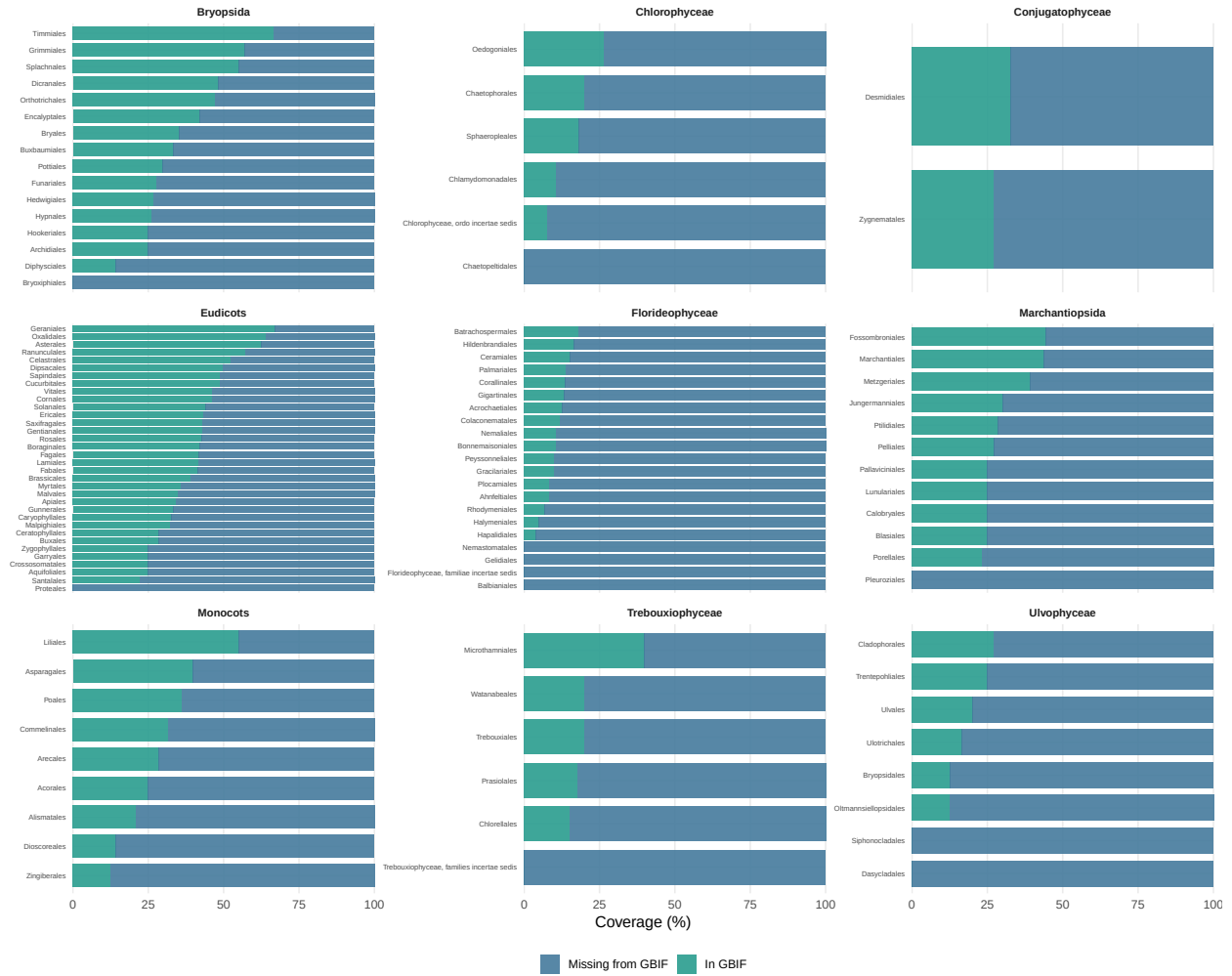


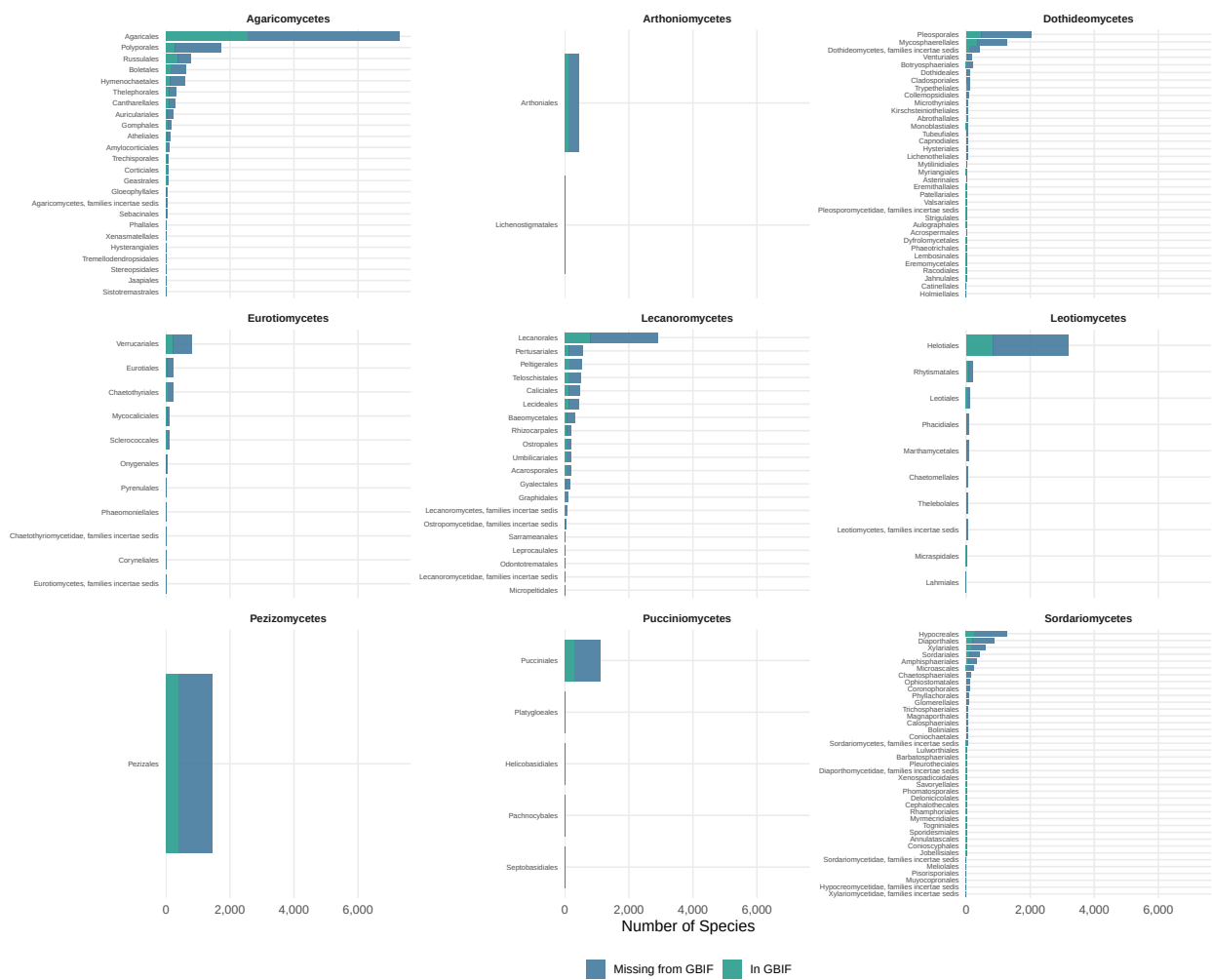
Table 2: Plantae: Species Coverage by Class

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Eudicots	34	11385	5479	5906	48.1
Monocots	9	2859	991	1868	34.7
Bryopsida	16	2150	738	1412	34.3
Florideophyceae	21	1271	167	1104	13.1
Chlorophyceae	6	1179	204	975	17.3
Marchantiopsida	12	939	289	650	30.8
Conjugatophyceae	2	822	260	562	31.6
Ulvophyceae	8	459	89	370	19.4
Trebouxiphyceae	6	329	50	279	15.2
Polypodiopsida	5	268	53	215	19.8
Pinopsida	3	184	83	101	45.1
Sphagnopsida	1	89	49	40	55.1
Magnoliids	4	88	18	70	20.5
Lycopodiopsida	3	75	16	59	21.3
Charophyceae	1	67	36	31	53.7
Bangiophyceae	2	61	7	54	11.5

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Polytrichopsida	1	59	27	32	45.8
Pyramimonadophyceae	2	58	14	44	24.1
Equisetopsida	1	39	10	29	25.6
Stylonematophyceae	1	33	3	30	9.1
Compsopogonophyceae	1	29	4	25	13.8
Chlorodendrophyceae	1	28	5	23	17.9
Ophioglossida	2	27	8	19	29.6
Klebsormidiophyceae	1	24	3	21	12.5
Mamiellophyceae	3	23	5	18	21.7
Andreaeopsida	1	19	8	11	42.1
Pedinophyceae	3	19	0	19	0.0
Coleochaetophyceae	2	18	6	12	33.3
Cyanidiophyceae	2	16	0	16	0.0
Anthocerotopsida	2	11	4	7	36.4
Tetraphidopsida	1	10	4	6	40.0
Glaucophyceae	2	9	2	7	22.2
Chlorophyta, classis incertae sedis	1	8	0	8	0.0
Nephrophyceae	1	8	3	5	37.5
Porphyridiophyceae	1	8	1	7	12.5
Rhodellophyceae	2	8	1	7	12.5
Ginkgoopsida	1	4	1	3	25.0
Mesostigmatophyceae	1	4	0	4	0.0
Oedipodiopsida	1	3	1	2	33.3
Palmophyllophyceae	2	2	0	2	0.0

Fungi

Fungi: Species Coverage by Order (Absolute)
Bar length = species in backbone | Green = in GBIF | Blue = missing



Fungi: Species Coverage by Order (Percentage)

All orders = 100% of backbone species | Green = % in GBIF | Blue = % missing

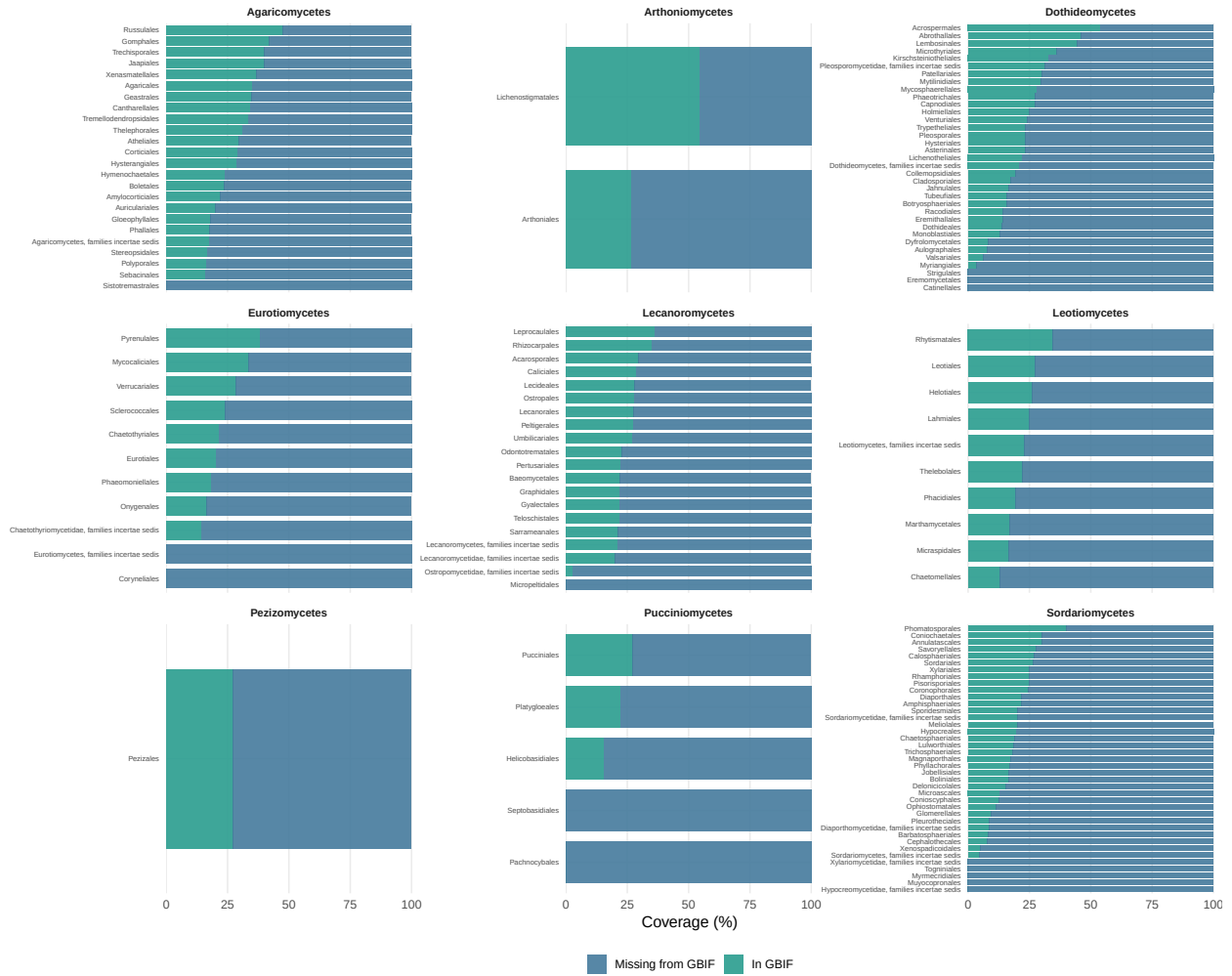


Table 3: Fungi: Species Coverage by Class

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Agaricomycetes	24	12746	4011	8735	31.5
Lecanoromycetes	20	6990	1840	5150	26.3
Dothideomycetes	35	5223	1241	3982	23.8
Sordariomycetes	39	4929	1008	3921	20.5
Leotiomyces	10	3852	998	2854	25.9
Eurotiomycetes	11	1576	404	1172	25.6
Pezizomycetes	1	1450	395	1055	27.2
Pucciniomycetes	5	1142	306	836	26.8
Arthoniomycetes	2	467	127	340	27.2
Ustilaginomycetes	2	433	114	319	26.3
Lichinomycetes	6	425	120	305	28.2
Laboulbeniomycetes	3	210	92	118	43.8
Ascomycota, ordines incertae sedis	1	206	27	179	13.1
Microbotryomycetes	5	196	42	154	21.4
Exobasidiomycetes	5	177	75	102	42.4
Tremellomycetes	6	176	51	125	29.0

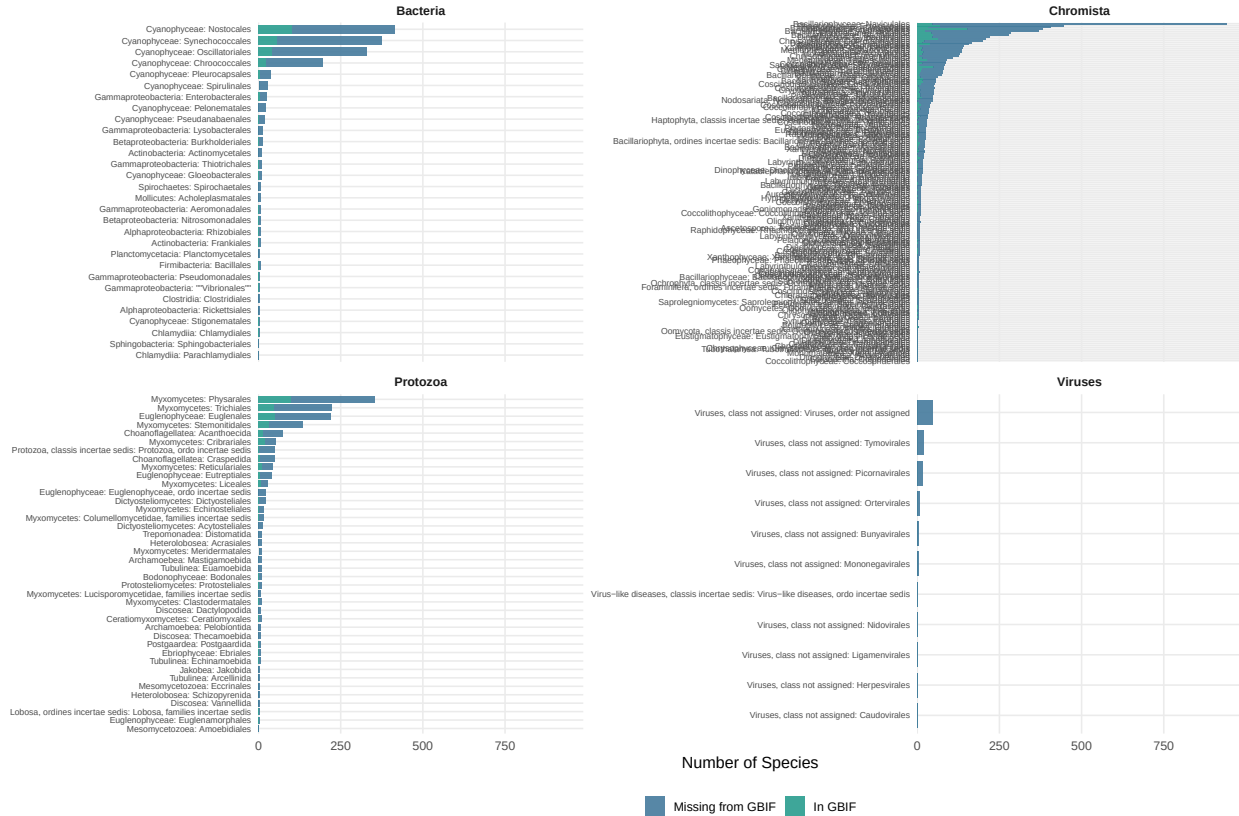
class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Microsporidiidea	2	164	3	161	1.8
Mucoromycetes	1	151	27	124	17.9
Dacrymycetes	1	114	30	84	26.3
Orbiliomycetes	1	97	25	72	25.8
Pezizomycotina, ordines incertae sedis	3	95	24	71	25.3
Taphrinomycetes	1	91	38	53	41.8
Entomophthoromycetes	1	73	12	61	16.4
Dipodascomycetes	1	64	4	60	6.2
Glomeromycetes	4	62	4	58	6.5
Chytridiomycetes	6	44	2	42	4.5
Pichiomyces	1	44	4	40	9.1
Physodermatomycetes	1	39	13	26	33.3
Rhizophyidiomycetes	1	32	2	30	6.2
Atractiellomycetes	1	31	3	28	9.7
Entorrhizomycetes	1	22	3	19	13.6
Umbelopsidomycetes	1	22	3	19	13.6
Malasseziomycetes	1	19	4	15	21.1
Synchytriomycetes	1	18	11	7	61.1
Saccharomycetes	2	17	1	16	5.9
Cystobasidiomycetes	2	14	1	13	7.1
Endogonomycetes	1	14	3	11	21.4
Blastocladiomycetes	3	13	0	13	0.0
Olpidiomycetes	1	13	0	13	0.0
Agaricostilbomycetes	1	12	2	10	16.7
Wallemiomycetes	1	12	2	10	16.7
Basidiomycota, ordines incertae sedis	1	10	0	10	0.0
Nelectomycetes	1	10	1	9	10.0
Pneumocystomycetes	1	9	0	9	0.0
Zoopagomycetes	1	9	0	9	0.0
Cladochytriomycetes	1	8	0	8	0.0
Rhizophlyctidomycetes	1	8	1	7	12.5
Rozellomycota, ordines incertae sedis	1	7	0	7	0.0
Spiculogloeomycetes	1	7	0	7	0.0
Archaeosporomycetes	1	6	0	6	0.0
Basidiobolomycetes	1	6	1	5	16.7
Lobulomycetes	1	6	1	5	16.7
Monoblepharidomycetes	1	6	0	6	0.0
Alloascoideomycetes	1	5	0	5	0.0
Archaeorhizomycetes	1	5	2	3	40.0
Kickxellomycetes	1	5	1	4	20.0
Paraglomeromycetes	1	5	0	5	0.0
Pucciniomycotina, ordines incerta sedis	1	5	1	4	20.0
Bartheletiomycetes	1	4	0	4	0.0
Hyaloraphidiomycetes	1	4	0	4	0.0
Neozygitomycetes	1	4	0	4	0.0
Rudimicrosporea	1	4	0	4	0.0
Schizosaccharomycetes	1	4	1	3	25.0
Neocallimastigomycetes	1	3	0	3	0.0

class	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Asellariomycetes	1	2	0	2	0.0

Other Kingdoms

Other Kingdoms: Species Coverage by Order

Bar length = species in backbone | Green = in GBIF | Blue = missing



Summary by Kingdom

Table 4: Species Coverage Summary by Kingdom

kingdom	n_orders	species_dyntaxa	species_gbif	species_missing	pct_coverage
Animalia	421	91009	30034	60975	33.0
Fungi	242	41651	11086	30565	26.6
Plantae	169	22692	8639	14053	38.1
Chromista	180	8040	1204	6836	15.0
Bacteria	30	1574	250	1324	15.9
Protozoa	40	1527	309	1218	20.2
Viruses	11	106	0	106	0.0

2. Threatened Species

This section analyzes coverage of threatened species from backbone in GBIF data.

Total taxa with threat status in backbone: 5,466

Found in GBIF: 4,657

Missing from GBIF: 809

Threatened Species Coverage in GBIF

CR=Critically Endangered, EN=Endangered, VU=Vulnerable, NT=Near Threatened, LC=Least Concern

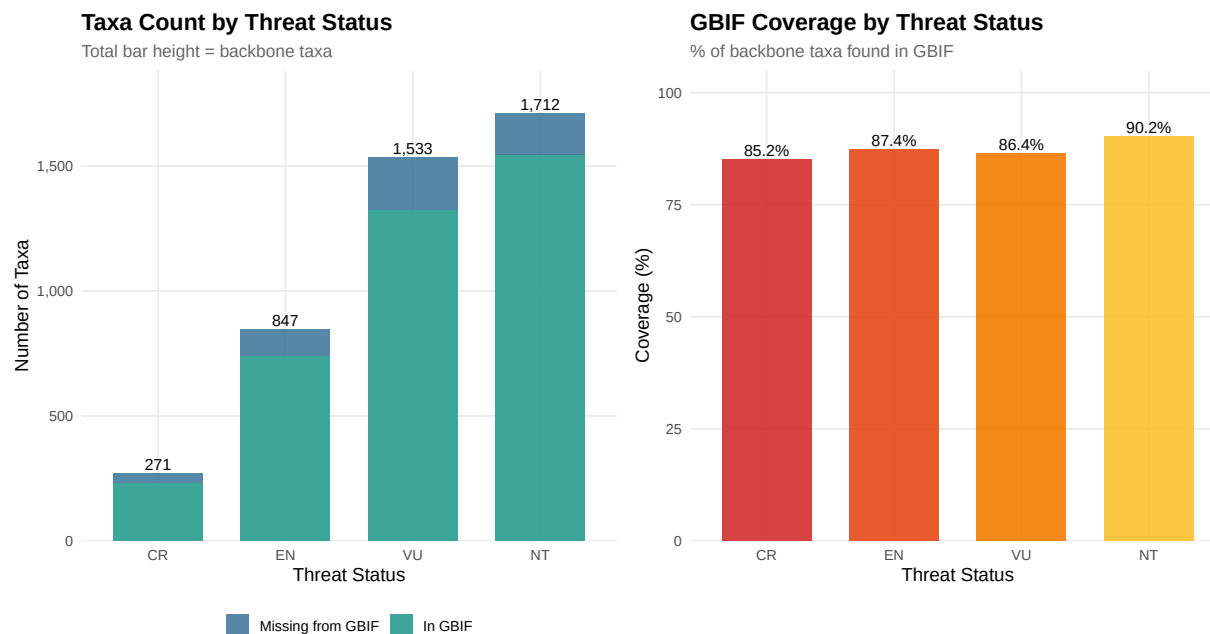


Table 5: Threatened Species Coverage Summary

Threat Status	In Backbone	In GBIF	Missing	Coverage %
CR	271	231	40	85.2
EN	847	740	107	87.4
NT	1712	1545	167	90.2
VU	1533	1324	209	86.4

523 threatened taxa (CR/EN/VU/NT) are missing from GBIF.



Complete List of Missing CR/EN Taxa

Total: 147 taxa (40 CR, 107 EN) ## Animalia (53 taxa) ### Arthropoda (43 taxa) #### Insecta (43 taxa) **Coleoptera** (5 taxa) *Bothrideridae* - *Teredus cylindricus* (CR)

Carabidae - *Calosoma auropunctatum* (CR) - *Callisthenes reticulatus* (EN)

Cerambycidae - *Monochamus sartor* (EN)

Tenebrionidae - *Hymenophorus doublieri* (EN)

Diptera (6 taxa) *Syrphidae* - *Arctophila bombiformis* (CR) - *Blera eoa* (EN) - *Caliprobola speciosa* (EN) - *Sphiximorpha subsestilis* (EN)

Asilidae - *Choerades lapponicus* (EN)

Dolichopodidae - *Systemus tener* (EN)

Hemiptera (3 taxa) *Pentatomidae* - *Aelia rostrata* (CR)

Microphysidae - *Loricula ruficeps* (EN)

Rhyparochromidae - *Peritrechus lundii* (EN)

Hymenoptera (9 taxa) *Andrenidae* - *Andrena morawitzi* (CR) - *Andrena bluethgeni* (EN)

Crabronidae - *Oxybelus latidens* (CR) - *Pemphredon fennica* (EN)

Halictidae - *Halictus quadricinctus* (CR) - *Sphecodes spinulosus* (CR)

Pompilidae - *Evagetes gibbulus* (CR)

Formicidae - *Leptothorax goesswaldi* (EN)

Ichneumonidae - *Ophion forticornis* (EN)

Lepidoptera (20 taxa) *Cosmopterigidae* - *Eteobalea tririvella* (CR)

Gelechiidae - *Dactylotula kinkerella* (CR) - *Syncopacma suecicella* (CR) - *Athrips amoenellus* (EN) - *Caryocolum arenariellum* (EN) - *Caryocolum cauligenellum* (EN) - *Chionodes violaceus* (EN) - *Syncopacma sangiella* (EN)

Glyphipterigidae - *Digitivalva valeriella* (CR)

Gracillariidae - *Phyllonorycter staintoniellus* (CR)

Nymphalidae - *Melitaea britomartis* (CR) - *Boloria polaris* (EN) - *Limenitis camilla* (EN)

Oecophoridae - *Epicallima formosella* (CR)

Autostichidae - *Holcopogon bubulcellus* (EN)

Coleophoridae - *Augasma aeratellum* (EN) - *Coleophora albella* (EN)

Crambidae - *Agriphila poliella* (EN)

Lycaenidae - *Lycaena helle* (EN)

Tineidae - *Nemapogon gliriellus* (EN)

Cnidaria (3 taxa)

Anthozoa (3 taxa) **Alcyonacea (2 taxa)** *Anthothelidae* - *Anthothela grandiflora* (CR)

Plexauridae - *Paramuricea placomus* (EN)

Scleractinia (1 taxa) *Caryophyllidae* - *Lophelia pertusa* (CR)

Mollusca (5 taxa)

Gastropoda (5 taxa) **Nudibranchia (1 taxa)** *Paracoryphellidae* - *Paracoryphella parva* (CR)

Patellogastropoda, families incertae sedis (1 taxa) *Patellidae* - *Patella vulgata* (CR)

Caenogastropoda, ordo incertae sedis (2 taxa) *Epitoniidae* - *Epitonium turtonis* (EN)

Newtoniellidae - *Eumetula arctica* (EN)

Cephalaspidea (1 taxa) *Philinidae* - *Philine punctata* (EN)

Annelida (1 taxa)

Polychaeta (1 taxa) **Terebellida (1 taxa)** *Ampharetidae* - *Lysippe fragilis* (EN)

Chordata (1 taxa)

Class unknown

- *Styela atlantica* (EN)

Fungi (51 taxa)

Ascomycota (14 taxa)

Arthoniomycetes (1 taxa) **Arthoniales (1 taxa)** *Arthoniaceae* - *Arthonia anomobrophila* (CR)

Lecanoromycetes (10 taxa) **Peltigerales** (3 taxa) *Lobariaceae* - *Lobaria amplissima* (CR) - *Lobaria virens* (EN)

Pannariaceae - *Moelleropsis nebulosa* (EN)

Rhizocarpales (1 taxa) *Rhizocarpaceae* - *Poeltinula interjecta* (CR)

Teloschistales (4 taxa) *Teloschistaceae* - *Calogaya lobulata* (CR) - *Caloplaca atroalba* (CR) - *Xanthomendoza fallax* (CR) - *Caloplaca ulcerosa* (EN)

Gyalectales (1 taxa) *Gyalectaceae* - *Gyalecta hypoleuca* (EN)

Lecanorales (1 taxa) *Stereocaulaceae* - *Squamarina degelii* (EN)

Dothideomycetes (1 taxa) **Strigulales** (1 taxa) *Strigulaceae* - *Strigula stigmatella* (EN)

Pezizomycetes (2 taxa) **Pezizales** (2 taxa) *Discinaceae* - *Gyromitra fastigiata* (EN)

Tarzettaceae - *Stephensia bombycina* (EN)

Basidiomycota (37 taxa)

Agaricomycetes (36 taxa) **Agaricales** (20 taxa) *Cortinariaceae* - *Cortinarius violaceipes* (CR) - *Cortinarius albertii* (EN) - *Cortinarius camptoros* (EN) - *Cortinarius citrinoolivaceus* (EN) - *Cortinarius majoranae* (EN) - *Cortinarius moëne-locozii* (EN) - *Cortinarius odoratus* (EN) - *Cortinarius parasuaveolens* (EN) - *Cortinarius prasinocyaneus* (EN) - *Cortinarius prasinus* (EN) - *Cortinarius pseudovulpinus* (EN) - *Cortinarius saporatus* (EN) - *Cortinarius suaveolens* (EN) - *Cortinarius vesterholtii* (EN) - *Cortinarius xanthochlorus* (EN) - *Cortinarius xanthosuavis* (EN)

Hygrophoraceae - *Hygrophorus calophyllus* (EN)

Lycoperdaceae - *Lycoperdon radicum* (EN)

Squamanitaceae - *Floccularia rickenii* (EN)

Tricholomataceae - *Porpoloma pes-caprae* (EN)

Polyporales (10 taxa) *Meruliaceae* - *Aurantiporus croceus* (CR) - *Sarcodontia crocea* (CR) - *Odoria alborubescens* (EN)

Polyporaceae - *Funalia trogii* (CR) - *Perenniporia fraxinea* (CR) - *Funalia gallica* (EN)

Steccherinaceae - *Antrodiella citrinella* (CR) - *Frantisekia mentschulensis* (EN)

Incrustoporiaceae - *Piloporia sajanensis* (EN) - *Skeletocutis jelicii* (EN)

Boletales (1 taxa) *Boletaceae* - *Xerocomus moravicus* (EN)

Gloeophyllales (1 taxa) *Gloeophyllaceae* - *Griseoporia carbonaria* (EN)

Gomphales (2 taxa) *Gomphaceae* - *Ramaria broomei* (EN) - *Ramaria echinovirens* (EN)

Hymenochaetales (1 taxa) *Hymenochaetaceae* - *Pelloporus triqueter* (EN)

Russulales (1 taxa) *Albatrellaceae* - *Albatrellus cristatus* (EN)

Dacrymycetes (1 taxa) **Dacrymycetales** (1 taxa) *Cerinomycetaceae* - *Dacrymyces enatus* (EN)

Plantae (43 taxa)

Bryophyta (11 taxa)

Bryopsida (11 taxa) **Pottiales** (4 taxa) *Pottiaceae* - *Tortula randii* (CR) - *Didymodon asperifolius* (EN)
- *Didymodon sinuosus* (EN) - *Tortella* × *cuspidatissima* (EN)

Bryales (4 taxa) *Bryaceae* - *Bryum funkii* (EN) - *Bryum knowltonii* (EN) - *Bryum marratii* (EN) - *Bryum wrightii* (EN)

Dicranales (1 taxa) *Ditrichaceae* - *Pleurodium palustre* (EN)

Grimmiales (1 taxa) *Seligeriaceae* - *Seligeria campylopoda* (EN)

Hypnales (1 taxa) *Plagiotheciaceae* - *Isopterygiopsis alpicola* (EN)

Tracheophyta (32 taxa)

Eudicots (30 taxa) **Asterales** (21 taxa) *Asteraceae* - Hieracium femsioëense (CR) - Hieracium toerense (CR) - Taraxacum bavaricum (CR) - Taraxacum tortilobum (CR) - Hieracium cacuminatum (EN) - Hieracium carcarophyllum (EN) - Hieracium haboëense (EN) - Hieracium hastato-ovatum (EN) - Hieracium helenae (EN) - Hieracium imbricatum (EN) - Hieracium maculosiforme (EN) - Hieracium marginelliceps (EN) - Hieracium oreades (EN) - Hieracium planifrons (EN) - Hieracium pseudonosmoides (EN) - Hieracium pycnodon (EN) - Hieracium remanentiforme (EN) - Pentanema conyzae (EN) - Pseudognaphalium luteoalbum (EN) - Taraxacum biformatum (EN) - Taraxacum dilutisqueameum (EN)

Lamiales (2 taxa) *Orobanchaceae* - *Orobanche purpurea* (CR)

Lamiaceae - *Mentha* × *gracilis* (EN)

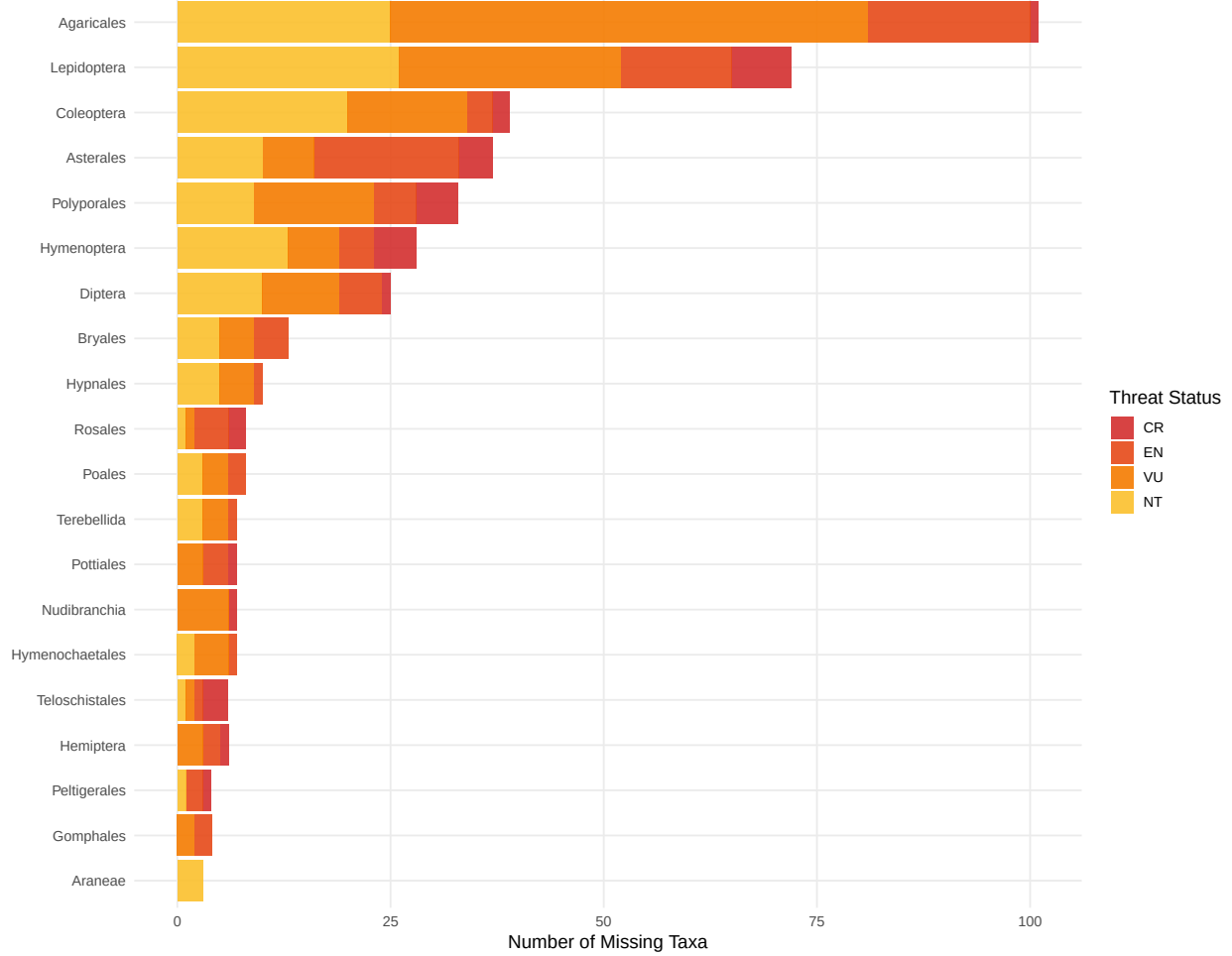
Rosales (6 taxa) *Rosaceae* - *Rubus gillottii* (CR) - *Rubus vigorosus* (CR) - *Rosa* $\times$ *inodora* (EN) - *Rubus* *onsalaënsis* (EN) - *Rubus pyramidalis* (EN) - *Sorbus obtusifolia* (EN)

Caryophyllales (1 taxa) *Caryophyllaceae* - *Gypsophila muralis* (EN)

Monocots (2 taxa) **Poales (2 taxa)** *Poaceae* - *Dactylis polygama* (EN) - *Vulpia bromoides* (EN)

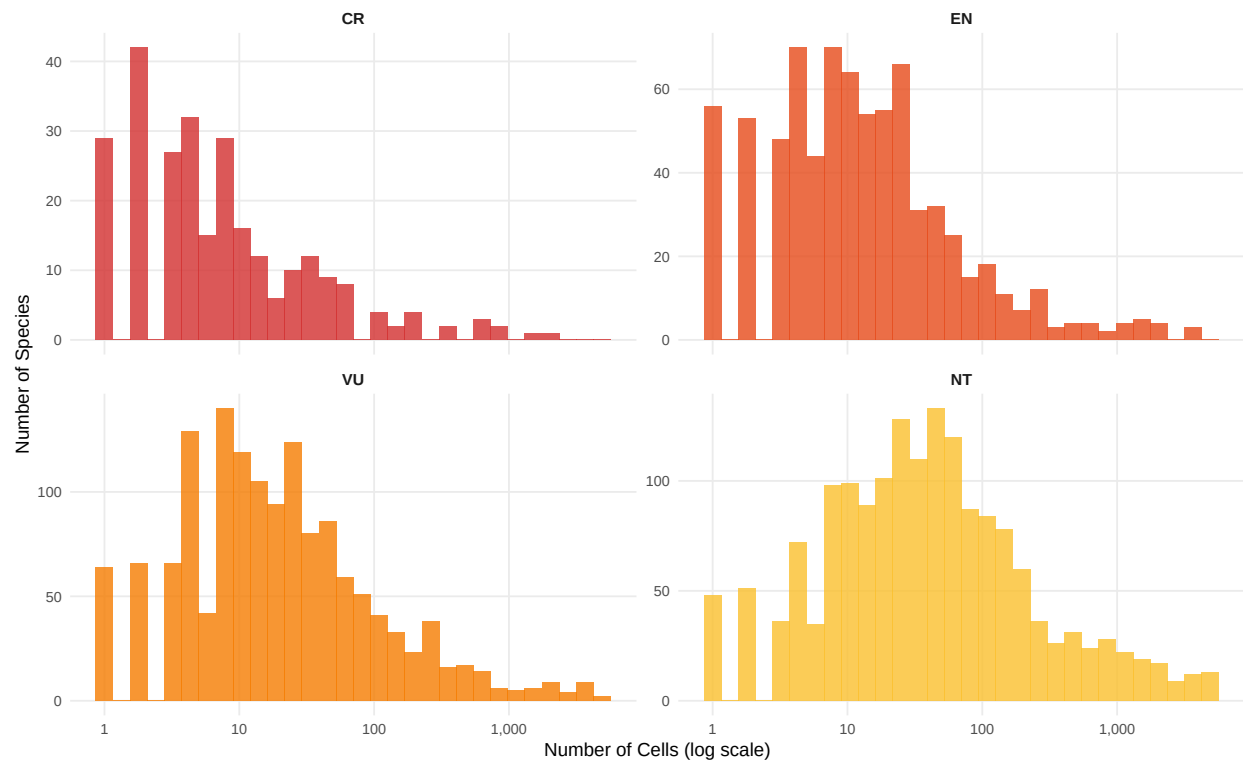
Missing Threatened Taxa by Order (Top 20)

Orders with most threatened species not in GBIF



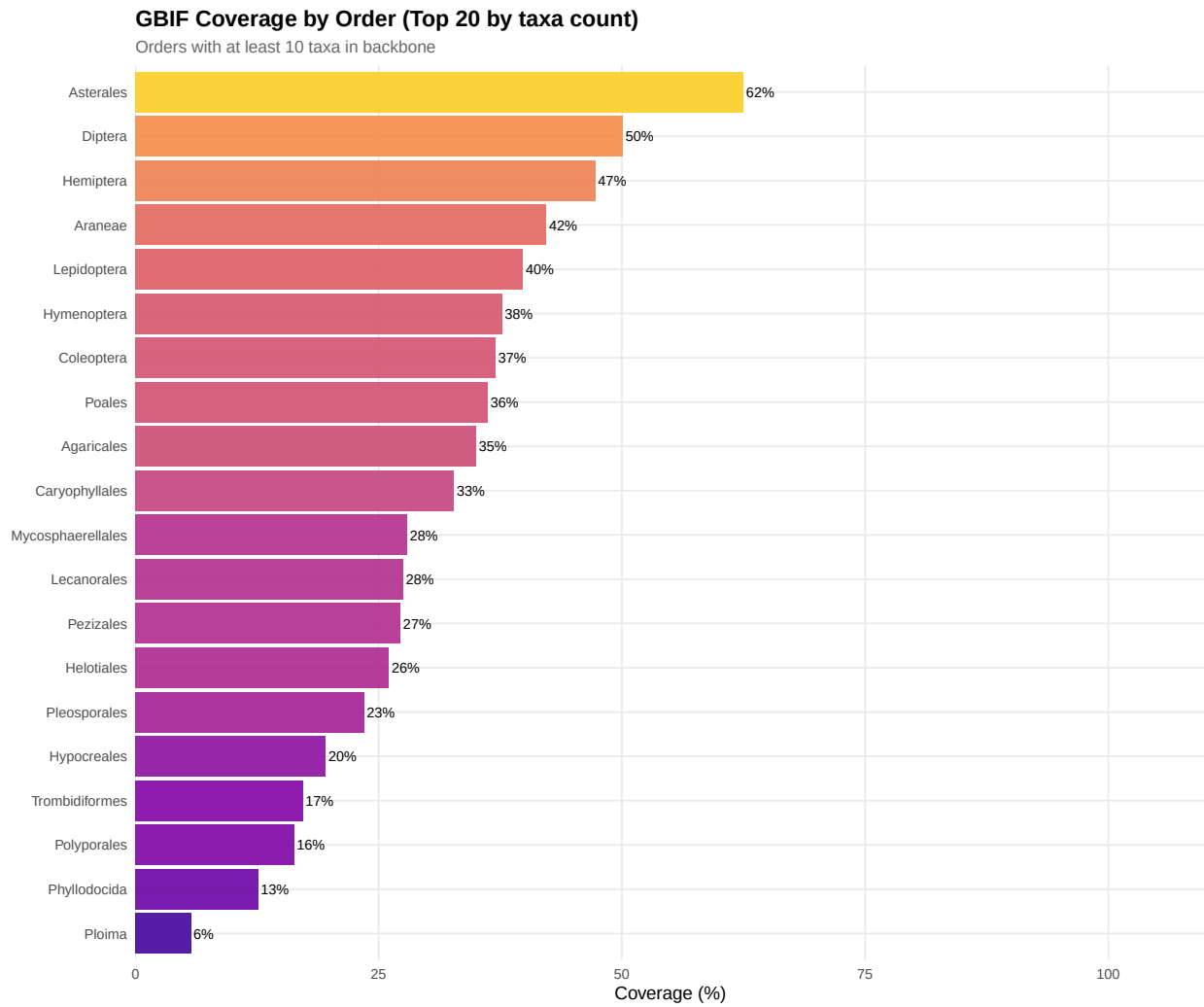
Spatial Coverage of Threatened Species in GBIF

Number of 10km grid cells with records

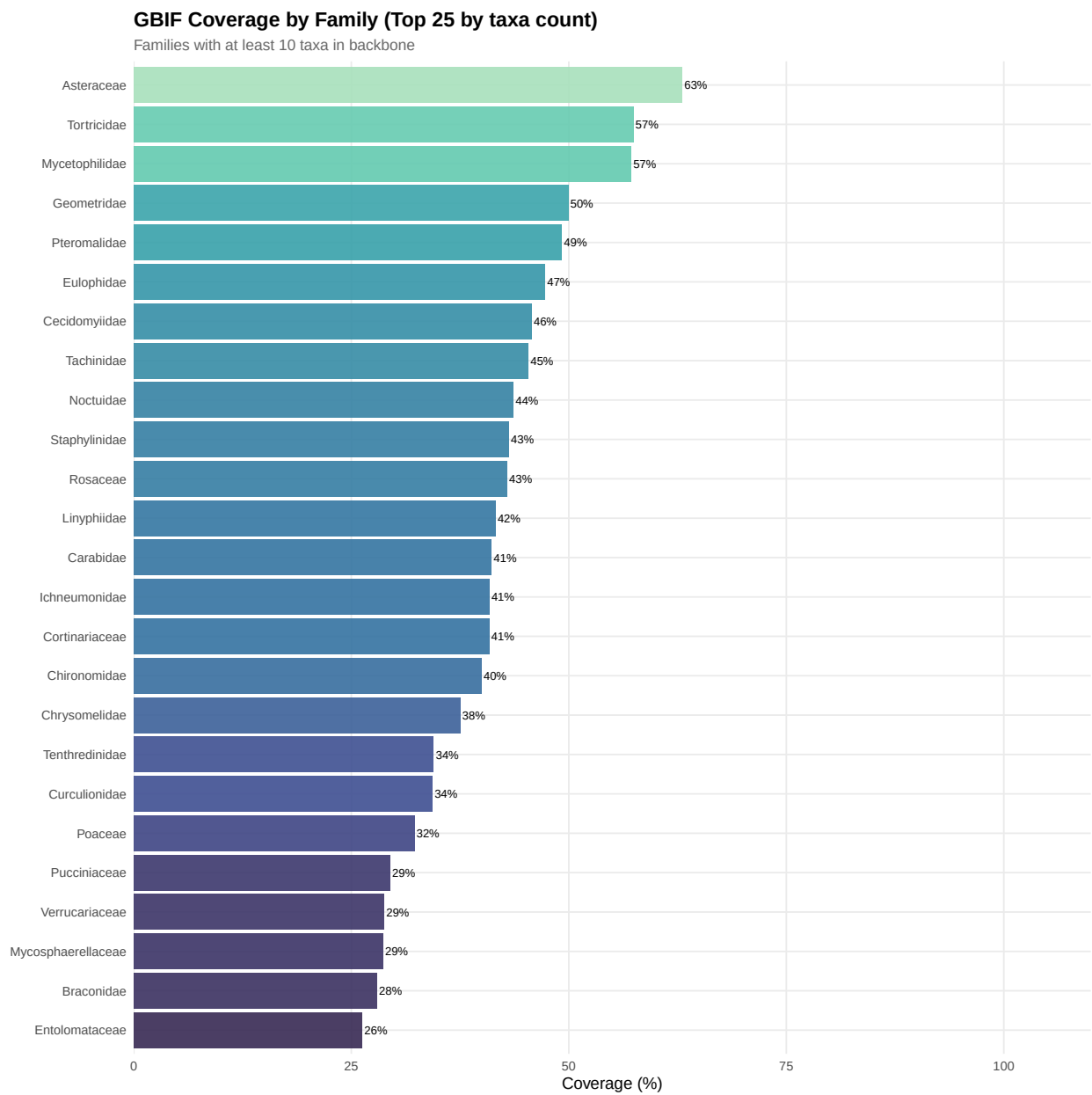


3. Coverage by Higher Taxonomy

3.1 Coverage by Order

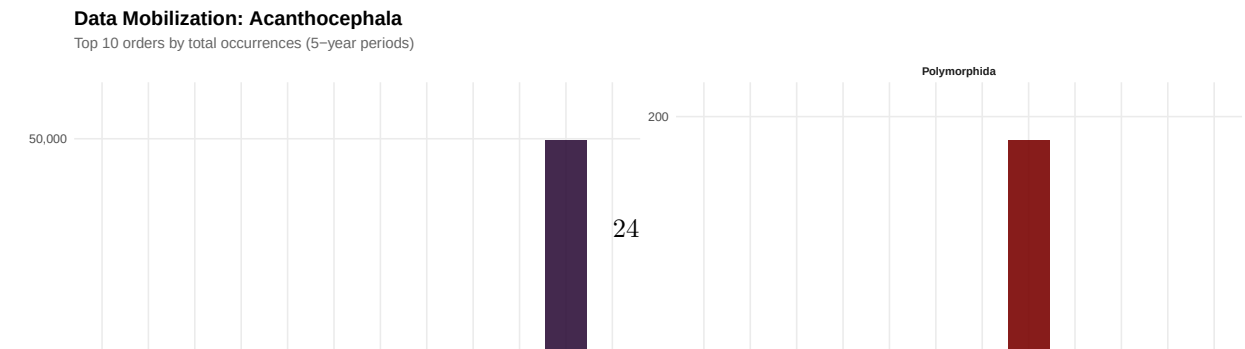
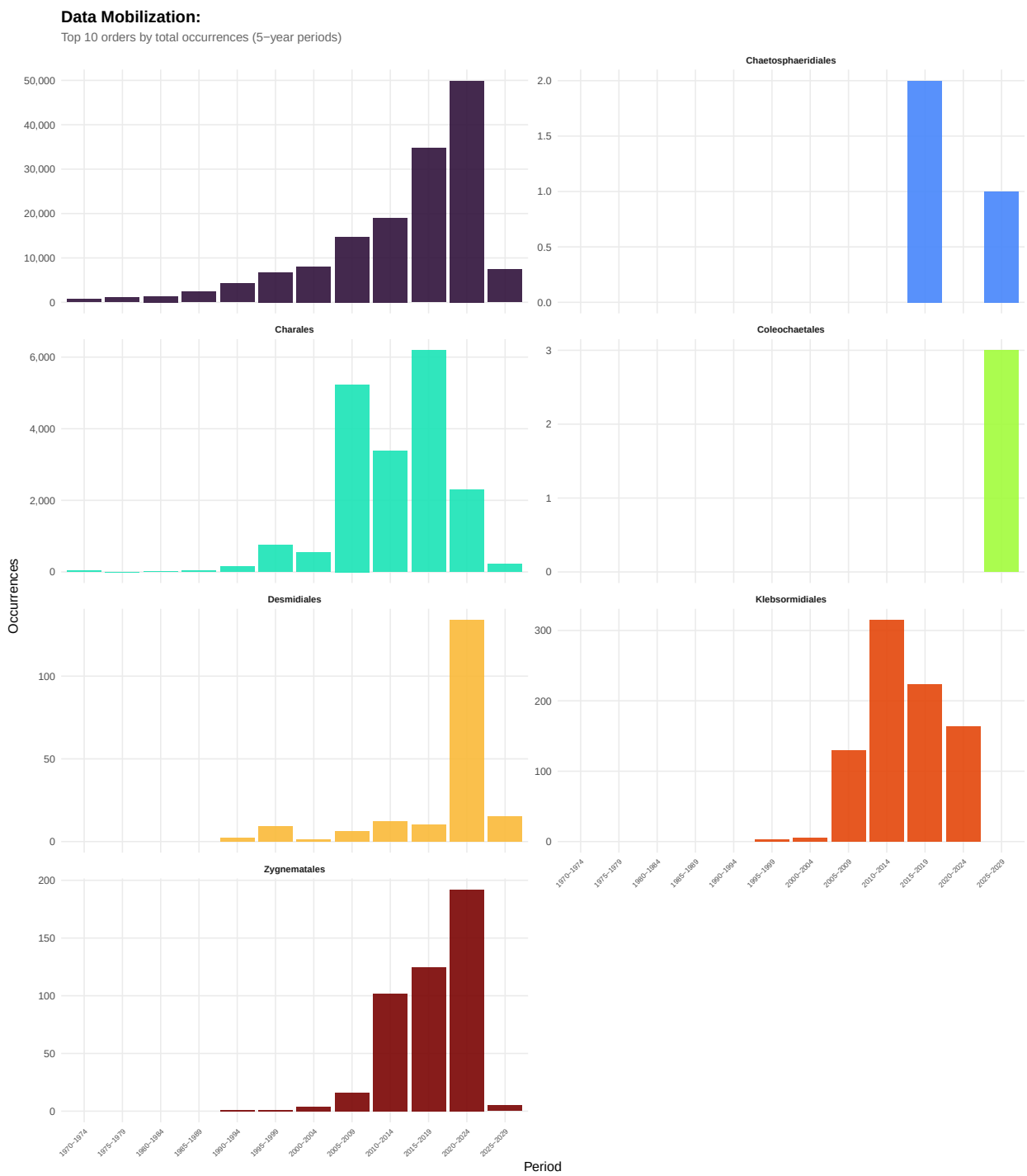


3.2 Coverage by Family

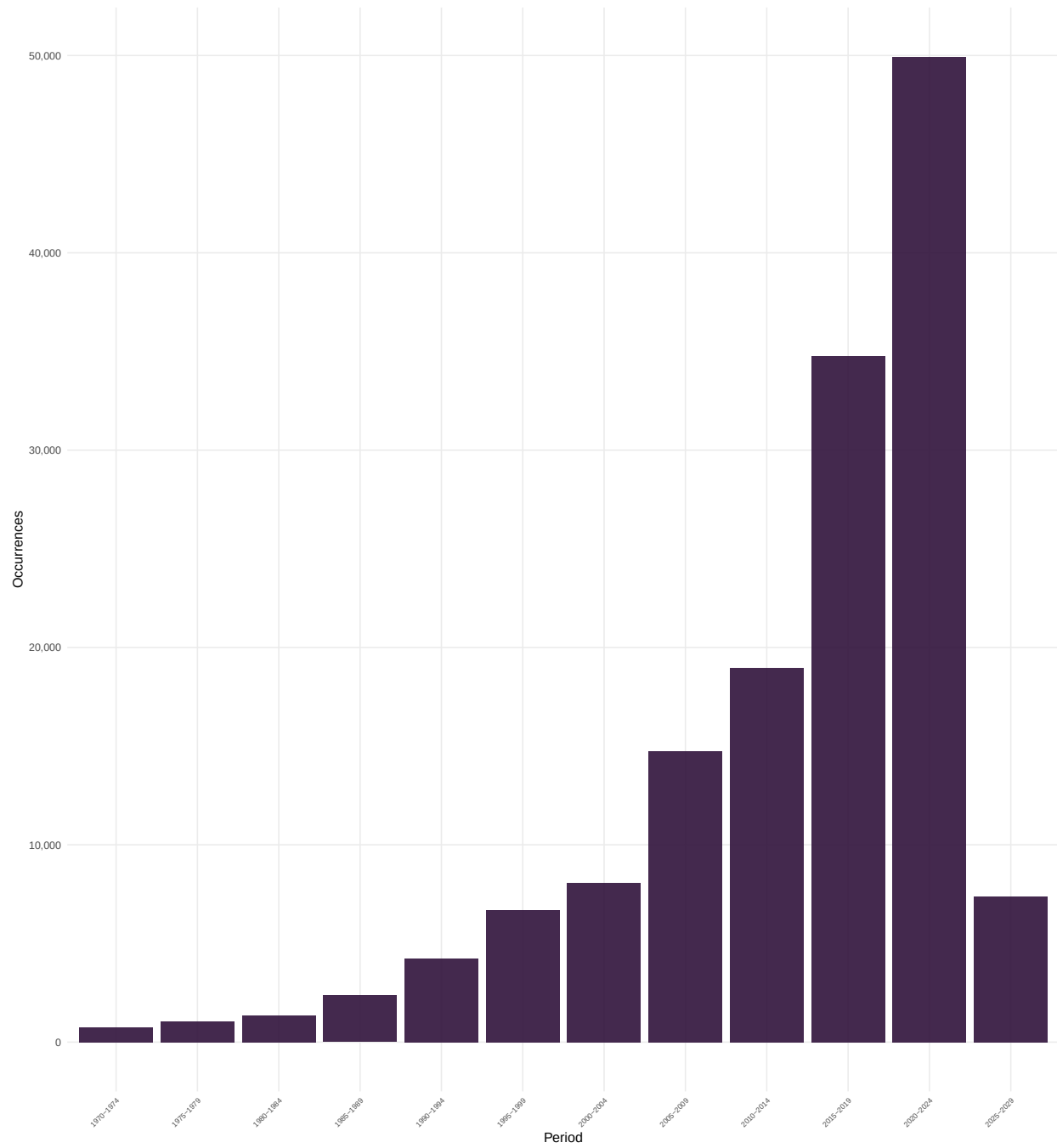


4. Data Mobilization Trends

4.1 Top 10 Orders by Phylum (5-Year Trends)

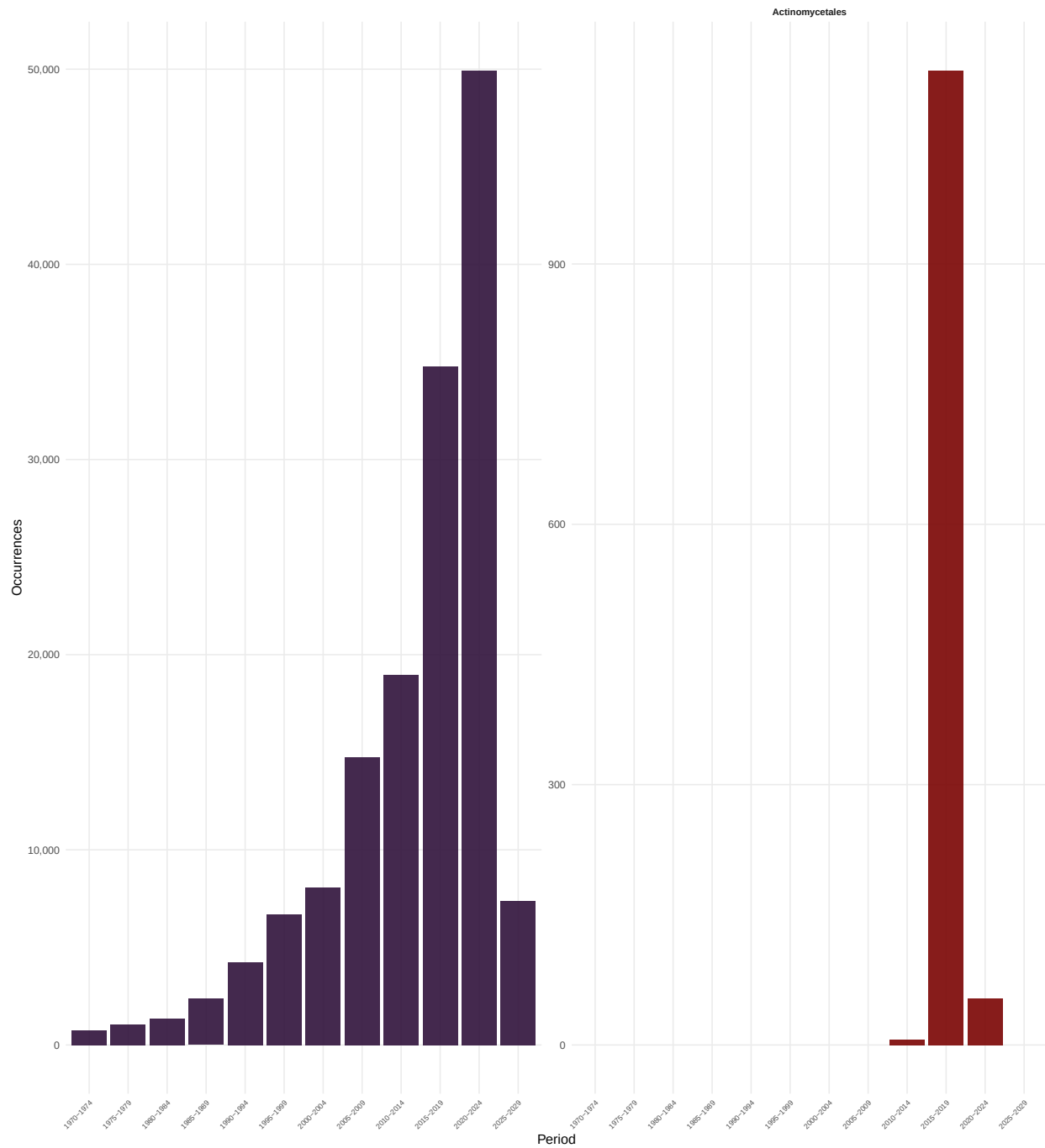


Data Mobilization: Acidobacteria
Top 10 orders by total occurrences (5-year periods)



Data Mobilization: Actinobacteria

Top 10 orders by total occurrences (5-year periods)

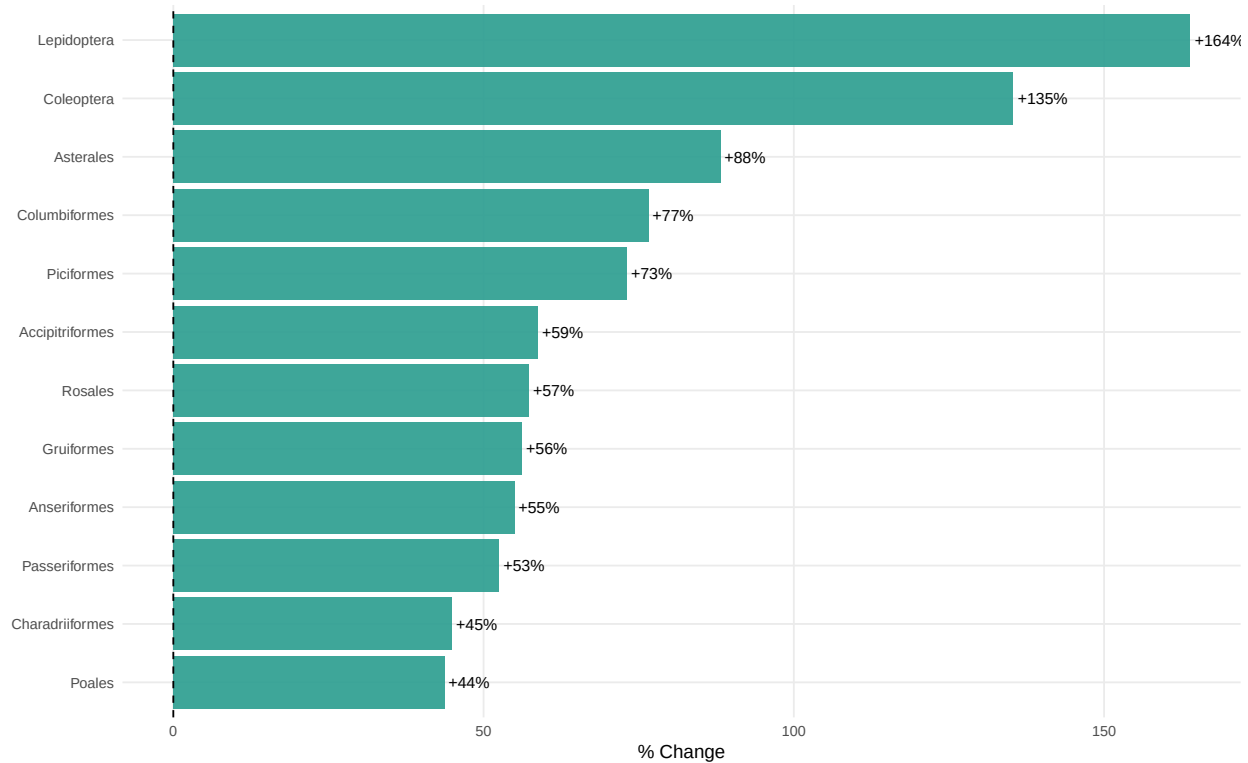


4.2 Recent vs Historical Change

Deriving top orders from temporal data...

Change in Data Mobilization by Order

Comparing 2016–2026 vs 2006–2015



5. Priority Taxa

Priority taxa are species that should be targeted for data collection efforts. These include: - Threatened species (CR, EN, VU, NT) missing from GBIF - Threatened species with poor spatial coverage in GBIF

5.1 Overview

```
## **1,351 taxa** require priority attention.
## - **CR**: 164 taxa
## - **EN**: 340 taxa
## - **VU**: 491 taxa
## - **NT**: 356 taxa
```

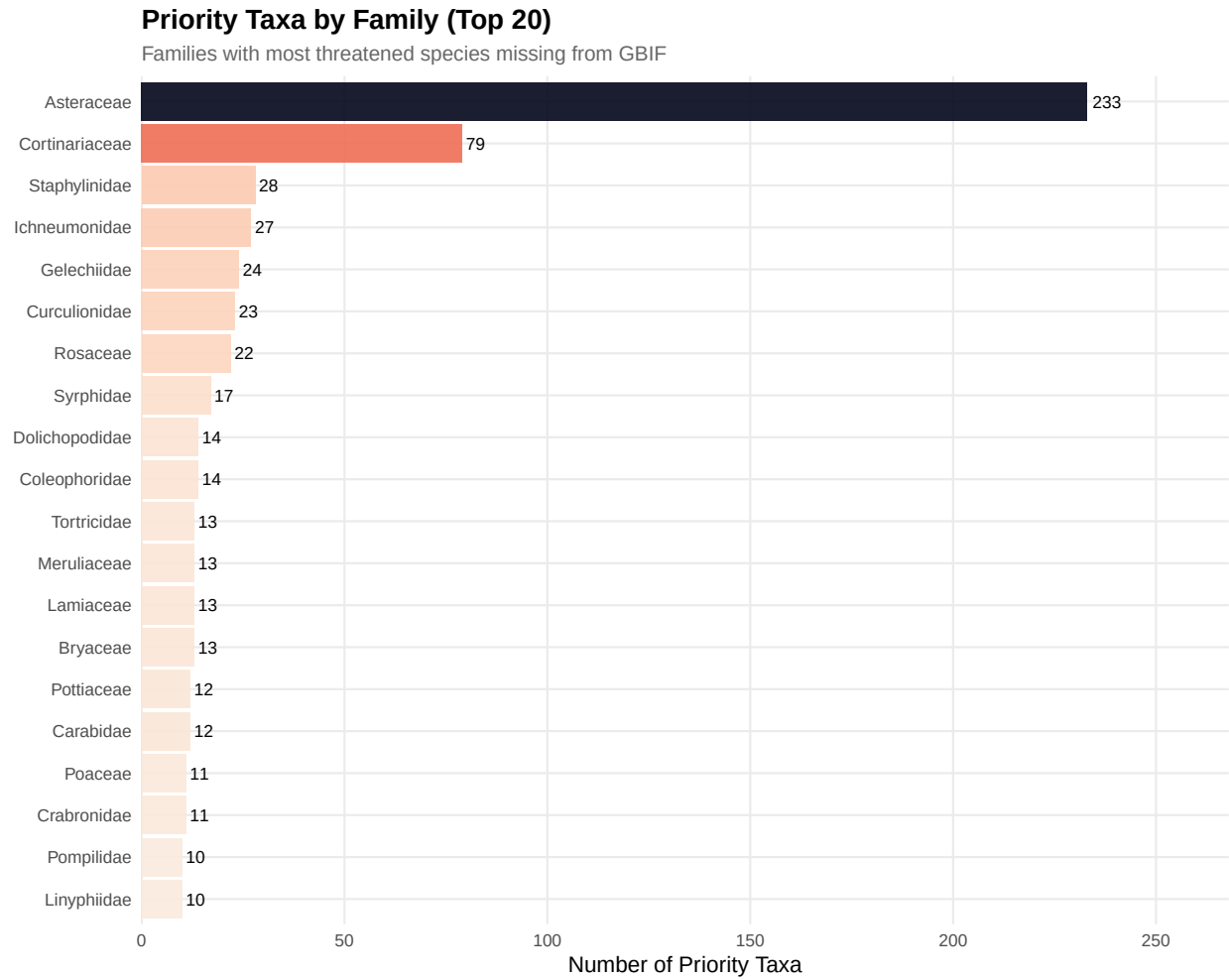
5.2 Highest Priority (CR/EN)

```
## **504 Critically Endangered and Endangered taxa** are priority for data mobilization.
```

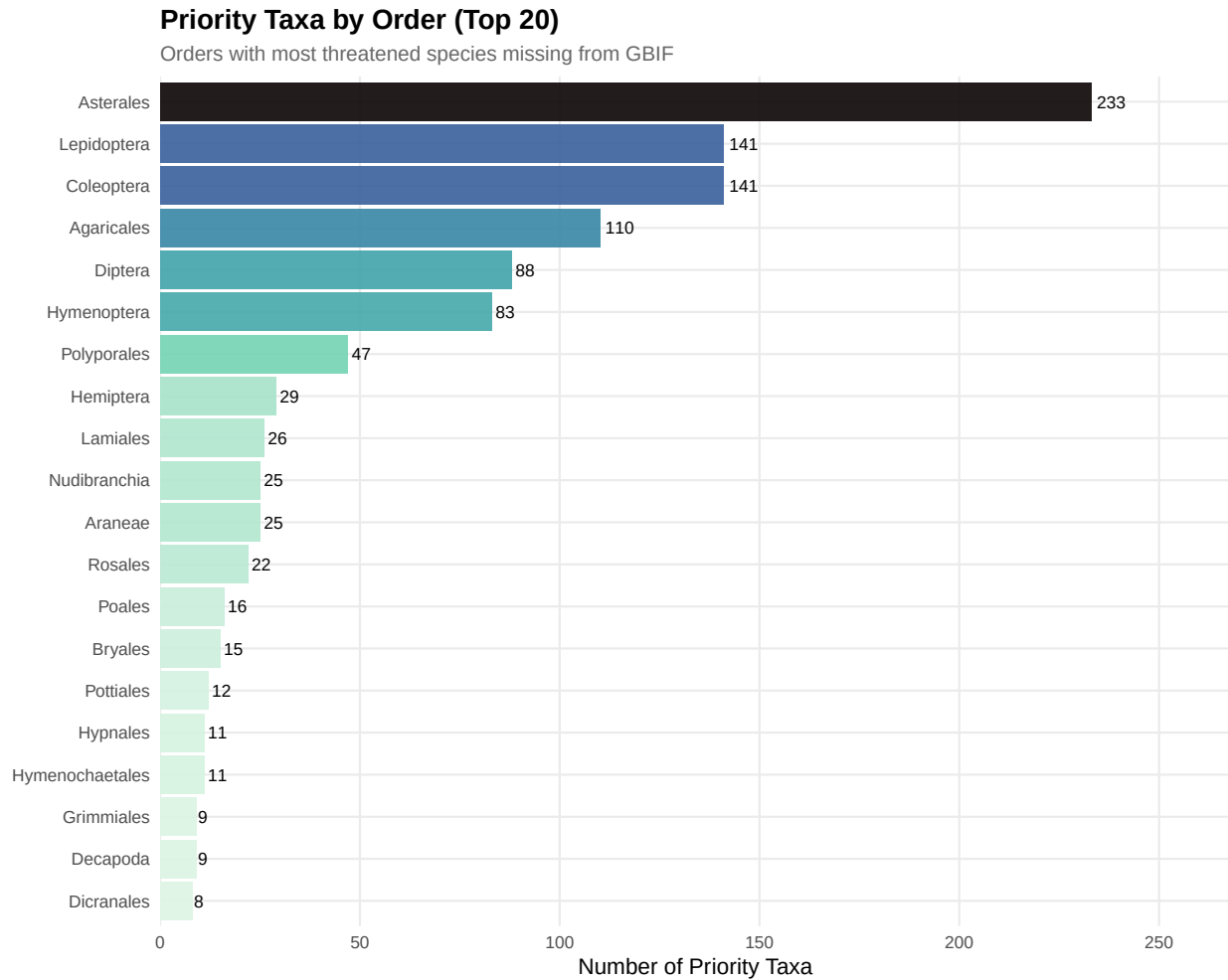
Table 6: Priority CR/EN Taxa (First 50)

scientificName	threatStatus	taxonRank	family	order
Andvakia parva	CR	species	Andvakiidae	Actiniaria
Octineon suecicum	CR	species	Octineonidae	Actiniaria
Cortinarius violaceipes	CR	species	Cortinariaceae	Agaricales
Anthothela grandiflora	CR	species	Anthothelidae	Alcyonacea
Alopecosa cursor	CR	species	Lycosidae	Araneae
Robertus insignis	CR	species	Theridiidae	Araneae
Arthonia anomobrophila	CR	species	Arthoniaceae	Arthoniales
Sporodophoron cretaceum	CR	species	Arthoniaceae	Arthoniales
Opegrapha cesareensis	CR	species	Opegraphaceae	Arthoniales
Hieracium abiegni	CR	species	Asteraceae	Asterales
Hieracium amoenifrons	CR	species	Asteraceae	Asterales
Hieracium concinnum	CR	species	Asteraceae	Asterales
Hieracium daniciforme	CR	species	Asteraceae	Asterales
Hieracium fensioëense	CR	species	Asteraceae	Asterales
Hieracium intermarginatum	CR	species	Asteraceae	Asterales
Hieracium ludificans	CR	species	Asteraceae	Asterales
Hieracium mallopodoides	CR	species	Asteraceae	Asterales
Hieracium manocolum	CR	species	Asteraceae	Asterales
Hieracium molucratum	CR	species	Asteraceae	Asterales
Hieracium mucrodentatum	CR	species	Asteraceae	Asterales
Hieracium multisigne	CR	species	Asteraceae	Asterales
Hieracium oligolepium	CR	species	Asteraceae	Asterales
Hieracium pachycalamum	CR	species	Asteraceae	Asterales
Hieracium parmiferum	CR	species	Asteraceae	Asterales
Hieracium placibile	CR	species	Asteraceae	Asterales
Hieracium psiloloma	CR	species	Asteraceae	Asterales
Hieracium psittacinum	CR	species	Asteraceae	Asterales
Hieracium semicanipes	CR	species	Asteraceae	Asterales
Hieracium serratoellipticum	CR	species	Asteraceae	Asterales
Hieracium spaniotrichum	CR	species	Asteraceae	Asterales
Hieracium steenhoffii	CR	species	Asteraceae	Asterales
Hieracium succulentifolium	CR	species	Asteraceae	Asterales
Hieracium toerense	CR	species	Asteraceae	Asterales
Taraxacum bavaricum	CR	species	Asteraceae	Asterales
Taraxacum lepidum	CR	species	Asteraceae	Asterales
Taraxacum obtusilobum	CR	species	Asteraceae	Asterales
Taraxacum plumbeum	CR	species	Asteraceae	Asterales
Taraxacum pruinatum	CR	species	Asteraceae	Asterales
Taraxacum subericinum	CR	species	Asteraceae	Asterales
Taraxacum tortilobum	CR	species	Asteraceae	Asterales
Trapeliopsis viridescens	CR	species	Trapeliaceae	Baeomycetales
Rinodina calcarea	CR	species	Physciaceae	Caliciales
Rinodina endophragmia	CR	species	Physciaceae	Caliciales
Rinodina fimbriata	CR	species	Physciaceae	Caliciales
Stellaria fennica	CR	species	Caryophyllaceae	Caryophyllales
Stellaria fennica	CR	species	Caryophyllaceae	Caryophyllales
Laona pruinosa	CR	species	Laonidae	Cephalaspidia
Chara filiformis	CR	species	Characeae	Charales
Plecotus austriacus	CR	species	Vespertilionidae	Chiroptera
Teredus cylindricus	CR	species	Bothrioderidae	Coleoptera

5.3 Priority by Family



5.4 Priority by Order



6. Summary & Recommendations

Key Findings

- **Overall coverage:** 30.7% (51,525 of 168,043 taxa)
- **Missing threatened taxa:** 523
 - CR: 40, EN: 107
- **Priority taxa:** 1,351

Recommendations

1. **Highest priority:** Focus on CR and EN species missing from GBIF

2. **Target underrepresented families** for systematic surveys
3. **Digitize museum specimens** for poorly covered taxonomic groups
4. **Maintain mobilization momentum** for orders showing growth
5. **Investigate declining orders** and address data collection gaps

Session Info

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## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/vecLib.framework/Versions/A/vecLib.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
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## [5] forcats_1.0.1     fstcore_0.10.0    shinyjs_2.1.1     cli_3.6.5
## [9] data.table_1.18.0 stringr_1.6.0      purrr_1.2.1       readr_2.1.6
## [13] here_1.0.2        lubridate_1.9.4    glue_1.8.0        viridis_0.6.5
## [17] viridisLite_0.4.2 scales_1.4.0       DT_0.34.0         sf_1.0-24
## [21] leaflet_2.2.3     plotly_4.12.0     ggplot2_4.0.1     tidyr_1.3.2
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##
## loaded via a namespace (and not attached):
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